Adarsh Raj

AI Engineer | Data & AI Engineer | Backend Engineer

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Professional Summary

Applied AI Engineer with 2.5+ years at GSK building production AI systems that combine LLMs, retrieval-augmented generation, and enterprise search on Azure AI and FastAPI. Comfortable owning backend systems end-to-end – from API design to Databricks-backed data pipelines – with a track record of double-digit gains in relevance, latency, and manual-effort reduction at enterprise scale.

Professional Experience

GSK (GlaxoSmithKline) — **Bangalore, India**

Data & AI Engineer — Jan 2024 – Present

Enterprise AI Platforms

- **Rover** – Enterprise AI Search & Knowledge Platform
- **Augmented Data Quality** – AI-Assisted Data Quality Automation Platform
- **Business Glossary Generator**

Core Engineering Contributions

- Indexed **76,000+** enterprise assets and lifted search relevance by **20–30%** by building Rover's multi-level semantic search and retrieval layer on **Azure AI Search** and **Azure OpenAI**.
- Cut manual data-quality rule-authoring effort by **70–80%** by designing an AI-assisted automation layer on **Azure OpenAI** and **FastAPI** that converts natural-language requirements into semantic mappings and SQL logic for Augmented Data Quality.
- Unified backend access across **Databricks Unity Catalog**, **ServiceNow**, **Collibra**, and **ARIS** by engineering **20–30** reusable **Python/FastAPI** REST APIs, load-tested to **70 requests/sec** in production.
- Gave governance teams dependency visibility across **10,000+** metadata assets by building a metadata lineage graph with **NetworkX** over Unity Catalog metadata.
- Kept enterprise AI platform releases stable and on schedule by serving as **Release Lead**, coordinating deployments, release validation, incident resolution, and production support.
- Sustained **20+** production deployments across **3+** enterprise integrations with materially fewer failures by architecting model-fallback logic and resilient service architecture across LLM-powered search and AI workflows.
- Cut API latency by **30–40%** and improved incident-detection time by modernizing backend observability with centralized logging, **Azure Monitor** alerting, caching, and production monitoring.

Technical Skills

Programming: Python, SQL, Java

AI / LLM: Azure OpenAI, Azure AI Search, RAG, LangChain, LangGraph, Semantic & Vector Search, Embeddings, Prompt Design, Search Ranking & Normalization

Backend: FastAPI, REST APIs, Pydantic, Async APIs

Data Engineering: Databricks, Spark SQL, Azure Data Factory, Unity Catalog

Cloud: Azure Functions, Docker, Azure DevOps

Databases: Azure SQL

Enterprise Platforms: Collibra, ServiceNow, ARIS, Ataccama

Monitoring: Azure Monitor, Application Insights, OpenTelemetry

Developer Tools: Git, JIRA, Power BI, NetworkX

Education

M.S. Ramaiah University of Applied Sciences — **Karnataka, India**

B.Tech, Computer Science & Information Science – CGPA: 8.2/10.0 — Aug 2020 – May 2024

Certifications & Awards

Awards: 2x Bronze Award (Metadata Platform & DevOps Automation; Supply Chain Warriors – Bronze Finalist, 44 teams) and multiple Quarterly Recognition Awards for enterprise AI innovation, metadata engineering, and DevOps automation at GSK.

Certifications: Generative AI – Microsoft | Responsible AI – Google | SQL – HackerRank | Python Masterclass – Udemy